

Visual Language Models (VLMs)



- P This is a picture of two teddy bears on the moon.
- What are they doing?
- P They are having a conversation.
- What object are they using?
- P It looks like a computer.
- Is this surprising?
- P Yes, it is surprising.
- Why is this picture surprising to you?
- I think it is surprising because teddy bears are not usually found on the moon.



- What is the common thing about these three images?
- P They are all flamingos.
- What is the difference between these three images?
- The first one is a cartoon, the second one is a real flamingo, and the third one is a 3D model of a flamingo.



- P This is an apple with a sticker on it.
- What does the sticker say?
- P The sticker says "iPod".
- Where is the photo taken?
- P It looks like it's taken in a backyard.
- Do you think it is printed or handwritten?
- P It looks like it's handwritten.
- What color is the sticker?
- P It's white.



- P This is a cityscape. It looks like Chicago.
- What makes you think this is Chicago?
- P I think it's Chicago because of the Shedd Aquarium in the background.



- What about this one? Which city is this and what famous landmark helped you recognise the city?
- P This is Tokyo. I think it's Tokyo because of the Tokyo Tower.

Plan

Tasks

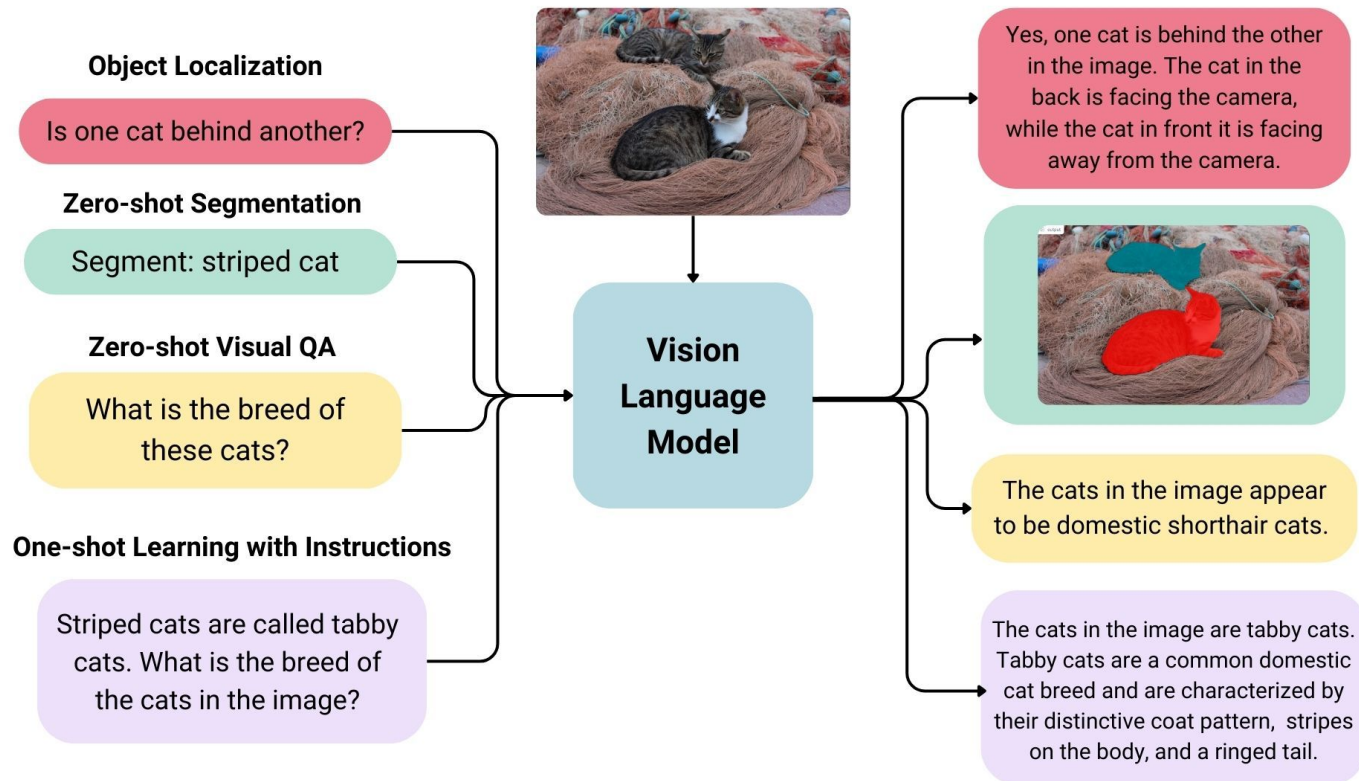
Review of previous

Architectures

Datasets

Leaderboards

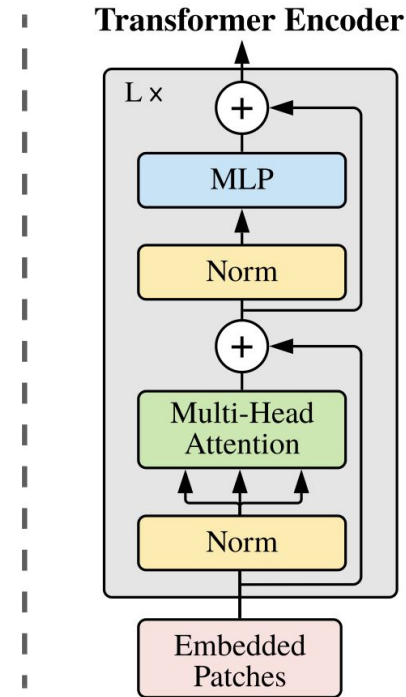
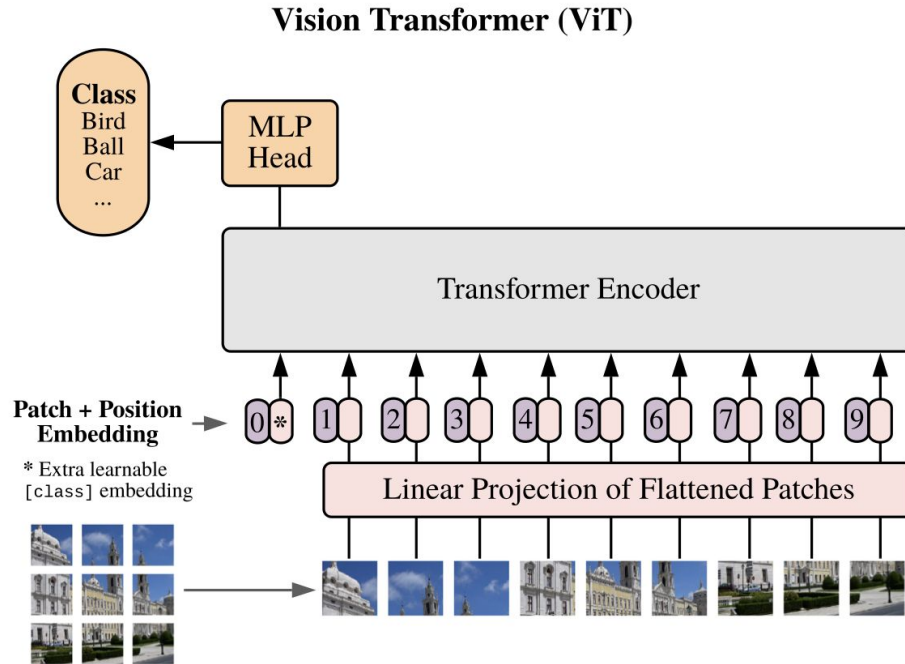
Tasks



What we know

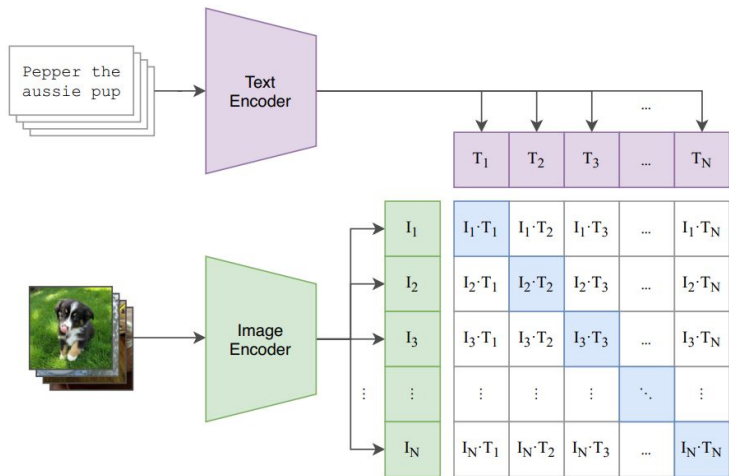
- Image convolution encoders
- Autoregressive language generation
- QKV attention mechanism

Vision Transformer (ViT) - Google, 2021

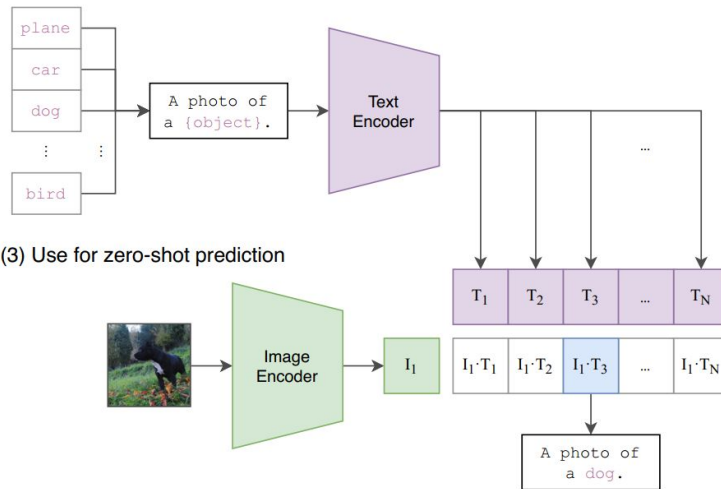


CLIP

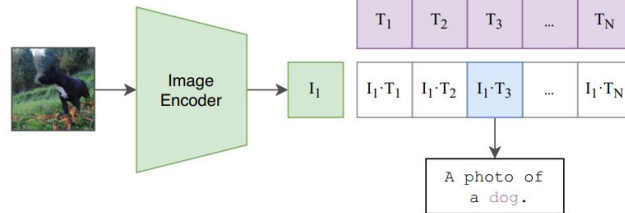
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



[CLIP paper](#)

CLIP. What for?

- Zero-shot classification
- Retrieval with extensions
- Image embedding standard

Caption RNN

```
class CaptionRNN(nn.Module):
    def __init__(self, image_embed_dim, embed_dim=256, hidden_dim=256, lstm_layers=1, dropout=0.1):
        super().__init__()
        self.cnn = MobileNet()

        self.lstm_layers = lstm_layers
        self.hidden_dim = hidden_dim

        self.image_embed_to_h0 = nn.Sequential(
            nn.Linear(image_embed_dim, lstm_layers * hidden_dim),
            nn.LeakyReLU(0.1),
        )
        self.image_embed_to_c0 = nn.Sequential(
            nn.Linear(image_embed_dim, lstm_layers * hidden_dim),
            nn.LeakyReLU(0.1),
        )

        self.embedding = nn.Embedding(
            num_embeddings=VOCAB_SIZE,
            embedding_dim=embed_dim,
            padding_idx=PAD_IDX
        )

        self.lstm = nn.LSTM(
            input_size=embed_dim,
            hidden_size=hidden_dim,
            num_layers=lstm_layers,
            batch_first=True
        )

        self.linear = nn.Sequential(
            nn.Dropout(dropout),
            nn.Linear(hidden_dim, VOCAB_SIZE)
        )
```

[Karpathy 2015 paper](#)

Flamingo - DeepMind 2022



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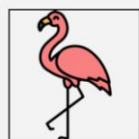
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Flamingo - DeepMind 2022

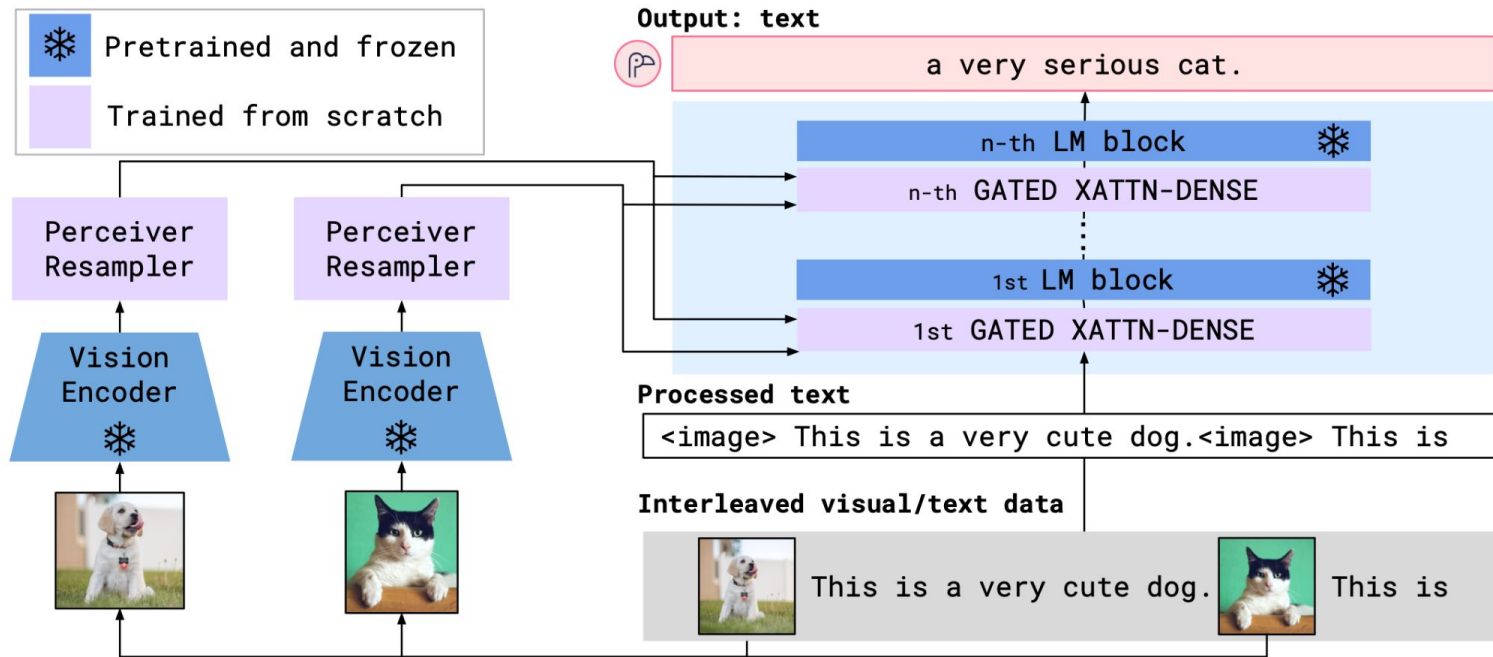
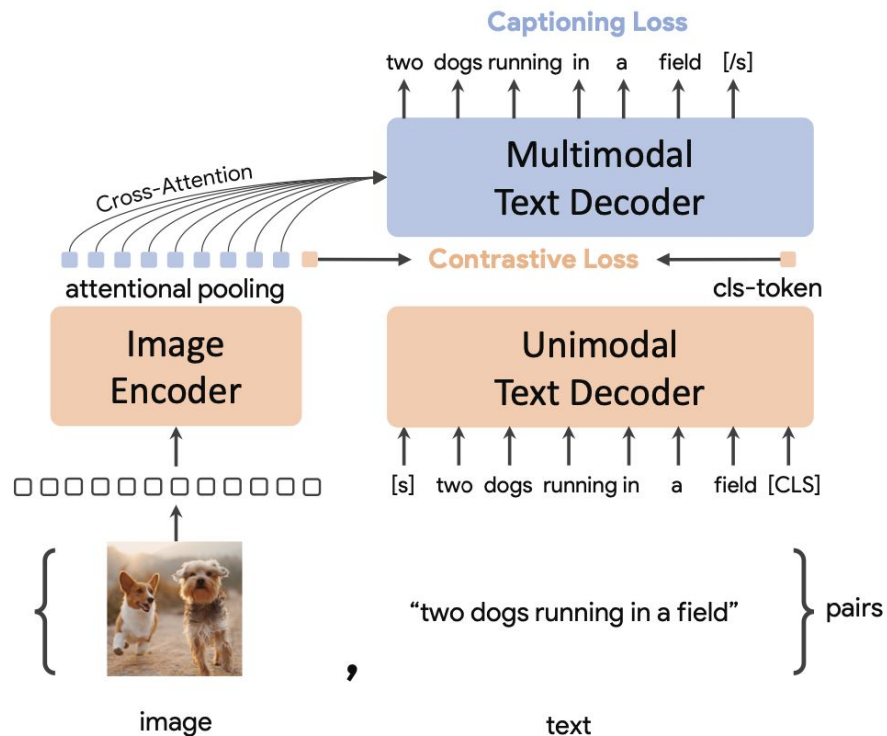


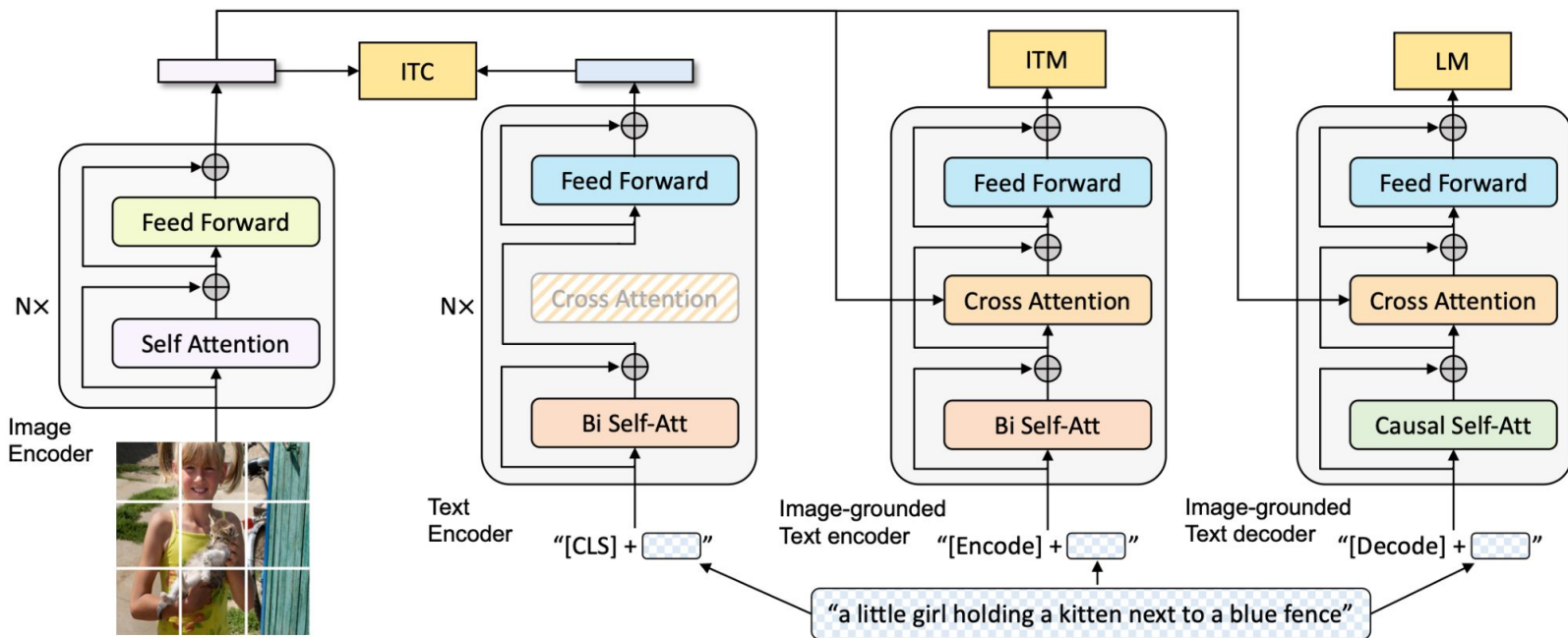
Figure 3: **Flamingo architecture overview.** Flamingo is a family of visual language models (VLMs) that take as input visual data interleaved with text and produce free-form text as output.

CoCa - Google 2022

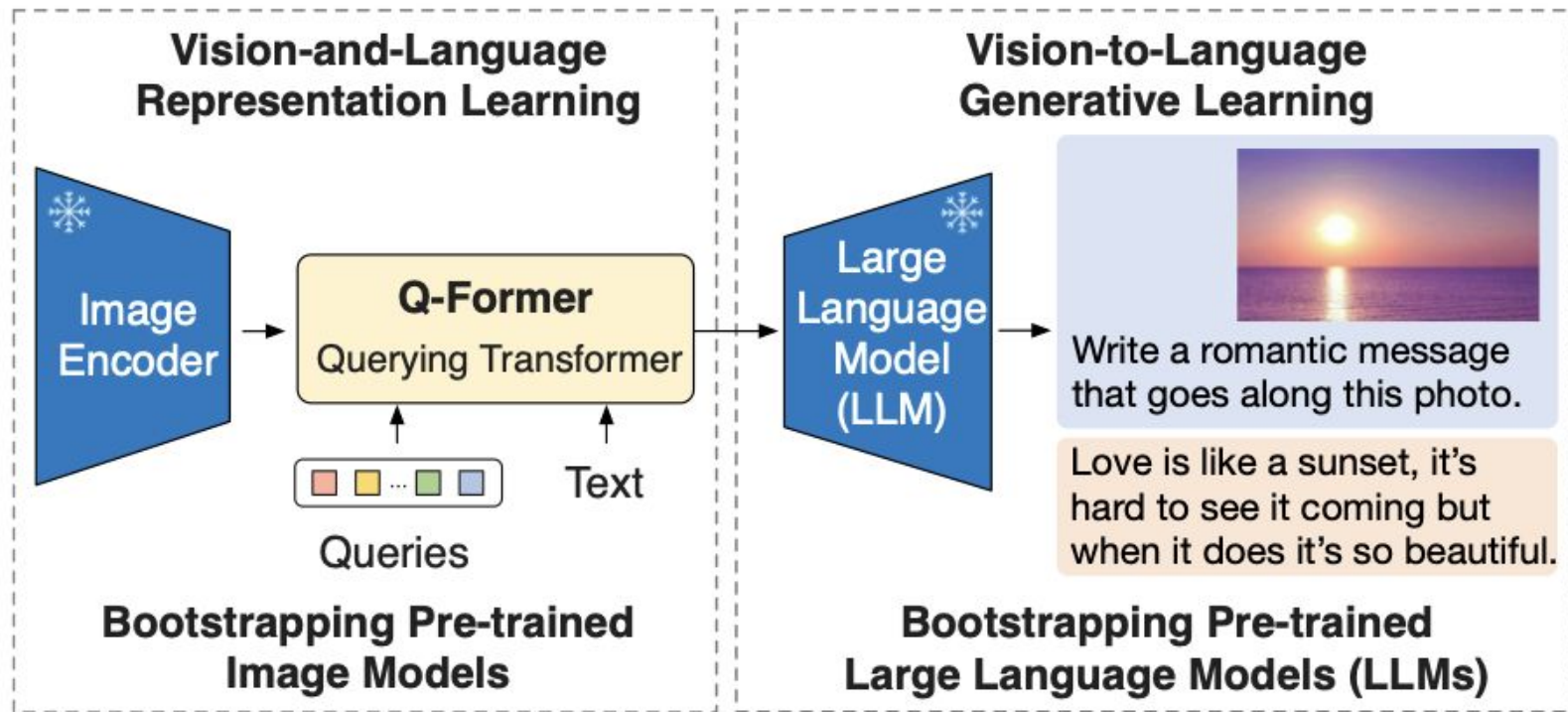


BLIP - Salesforce 2022

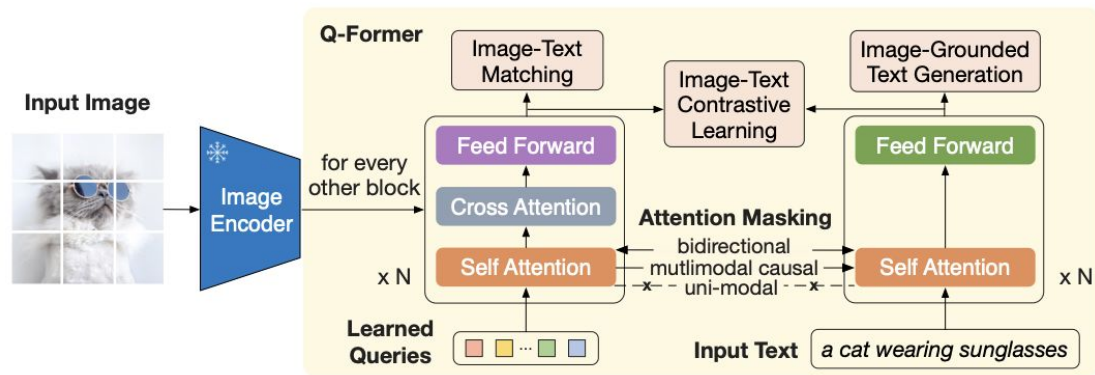
Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation



BLIP-2 - Salesforce 2023



BLIP-2 - Salesforce 2023



Q: query token positions; **T:** text token positions.

■ masked □ unmasked

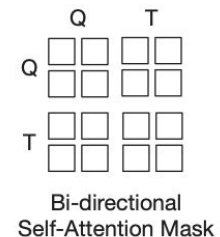


Image-Text Matching

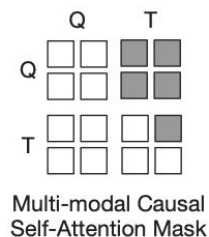


Image-Grounded Text Generation

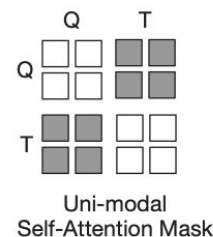
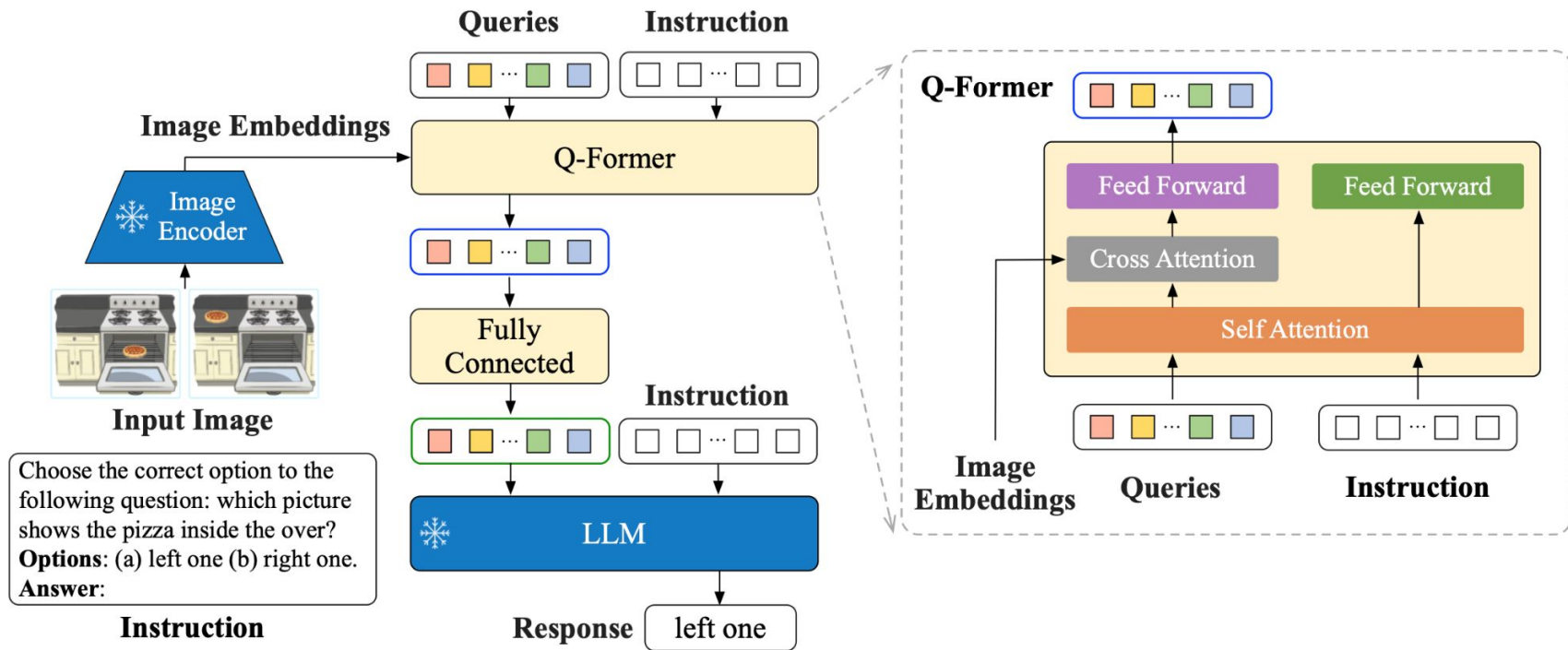
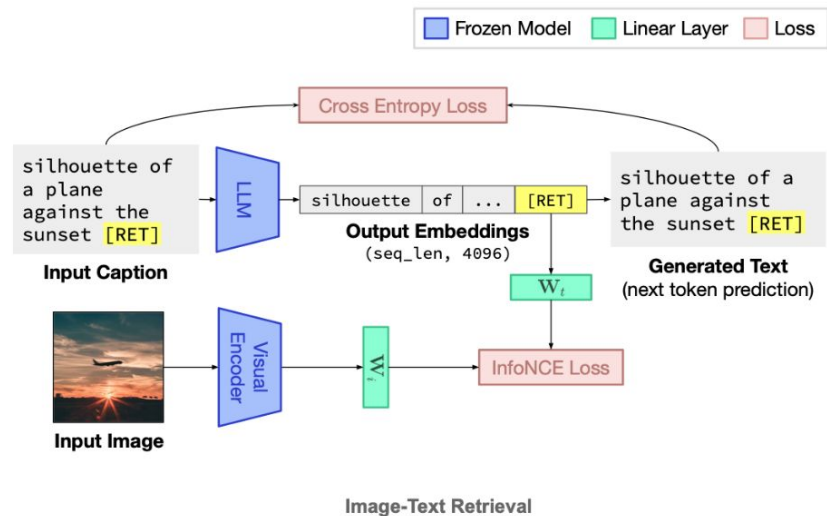
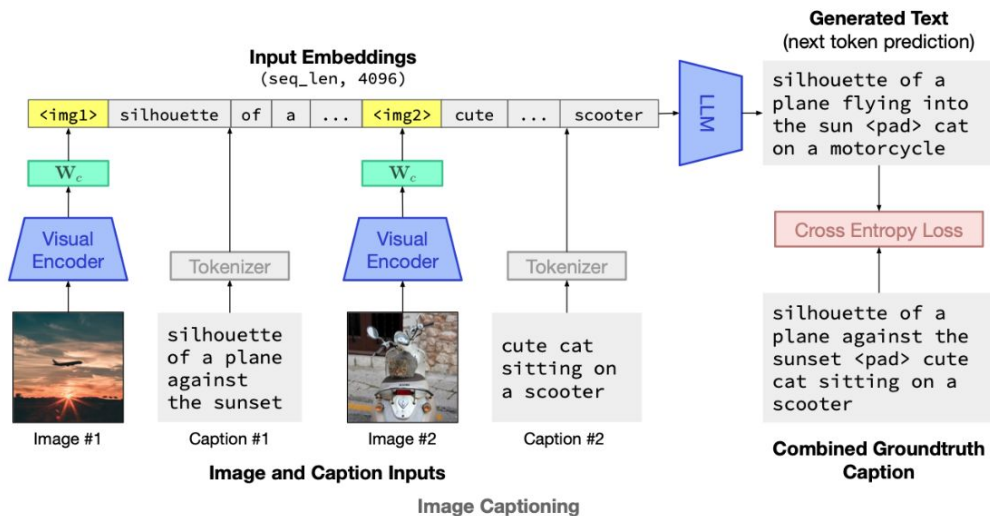


Image-Text Contrastive Learning

InstructBLIP - Salesforce 2023



FROMAGe - 2023



FROMAGe - 2023



What is this bird?



This is a male song sparrow (*Passer domesticus*) in a feeder. The bird is eating seeds.

What other birds can this feeder attract?



This feeder can attract many types of birds. The birds that can eat the seeds include the song sparrow, the bluebird, the titmouse, the black-headed hummingbird and the goldfinch.

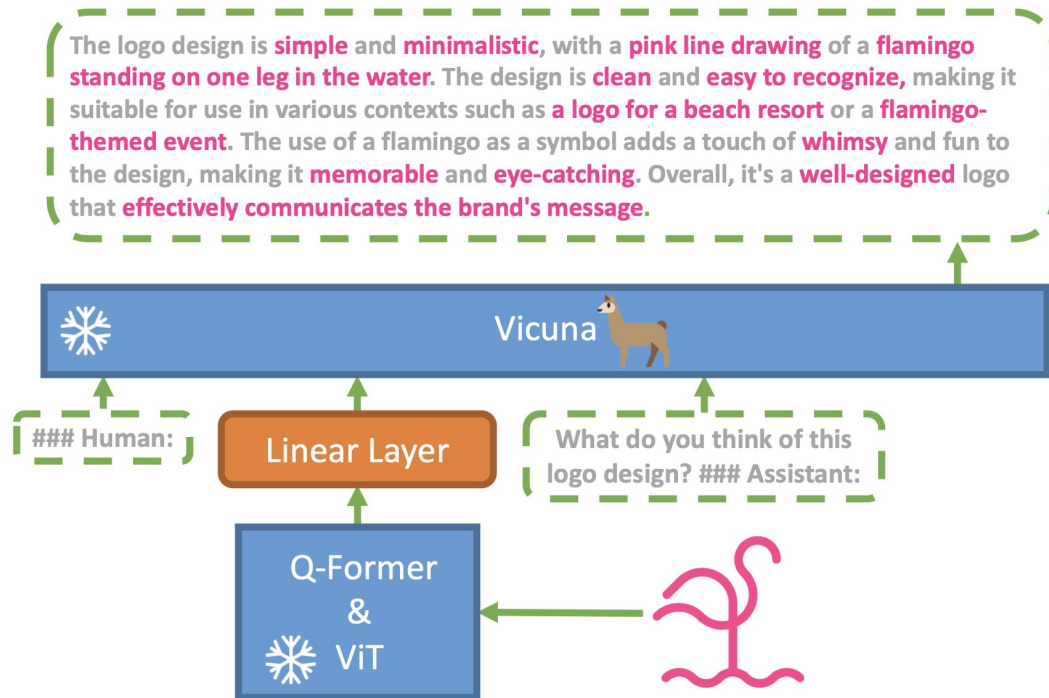
Show me pictures of some of these birds.



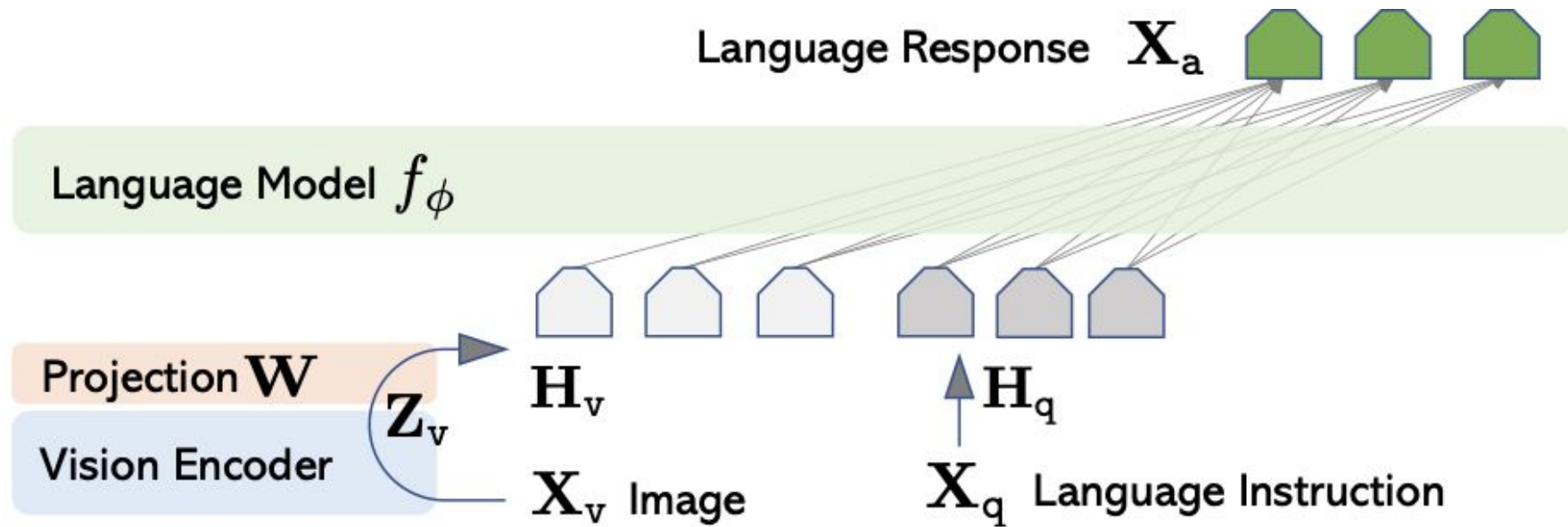
Here are pictures of some of the birds that can eat the seeds that this feeder attracts.



MiniGPT-4 - KAUST, 2023



LLaVA - Microsoft 2023

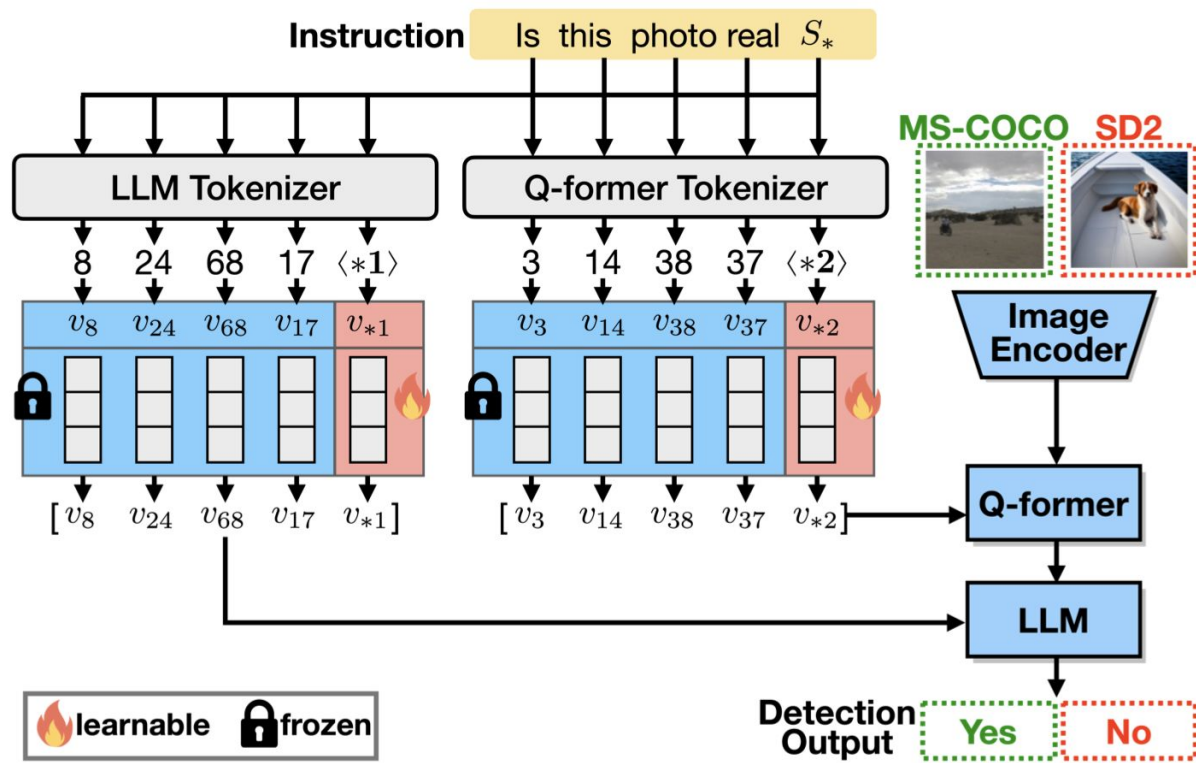


VLM research

Idefics2 - huggingface 2024 - [What matters when building vision-language models?](#)

MM1: [Methods, Analysis & Insights from Multimodal LLM Pre-training](#) - 2024, Apple

AntifakePrompt - Salesforce 2023



Leaderboard and datasets

https://huggingface.co/spaces/opencompass/open_vlm_leaderboard

https://github.com/jingyi0000/VLM_survey

https://huggingface.co/docs/transformers/en/tasks/document_question_answering